FLG (filaggrin, P20930) — Molecular Function: Keratin/IF Binding vs. Structural Molecule Activity

Focus: function_support · **Hypothesis slug:** mf-keratin-binding-vs-structural **Gene:** FLG / P20930 (Homo sapiens, NCBITaxon:9606) · **Term under test:** GO:0005198 structural molecule activity

> **Seed hypothesis:** The molecular function of the mature filaggrin repeat unit of human FLG (UniProt P20930) is *keratin intermediate filament binding that drives filament bundling*, not structural molecule activity.

Summary

Verdict: Supported, with one important qualifier. The immediate molecular function of the mature ~324-aa filaggrin repeat unit is best described as **intermediate-filament / keratin binding that drives keratin bundling** (GO:0019215 / GO:1990254), *not* a generic **structural molecule activity** (GO:0005198). This conclusion is reached convergently from an original biophysical/composition analysis of the P20930 sequence and from primary and review literature. Every repeat region examined (residues 258–306, 374–428, and the full 374–3872 tandem array) is intrinsically disordered, strongly hydrophilic, poor at forming both α-helix and β-sheet, Ser/His/Arg-rich, and cationic only at acidic skin pH. That biophysical profile is the signature of a disordered cationic polyelectrolyte that binds and condenses a *pre-formed* partner filament — the opposite of a rigid, ordered, load-bearing architectural subunit such as keratin, collagen, or tubulin.

The literature reinforces the same reading. Filaggrin is repeatedly defined as an intermediate-filament-associated protein that *functions to aggregate keratin intermediate filaments* (P 1429717) and was demonstrated in vitro to *aggregate keratin filaments specifically into bundles*

(P 6195345).
Crucially, filaggrin does not persist as an architectural element: it is progressively deiminated (charge-neutralized) and then completely proteolyzed to free amino acids / natural moisturizing factor (NMF) (P 33462753 P 23331681). A polypeptide that transiently condenses a partner network and is then destroyed is functionally a binding activity, not a lasting structural constituent.

The **qualifier** that keeps this from being a clean refutation of GO:0005198 is that a fraction of filaggrin is separately isopeptide (transglutaminase) cross-linked into the **cornified envelope** (CE; annotated CC GO:0001533, IDA). That covalently immobilized fraction *is* a bona-fide architectural contribution and is the legitimate basis for a structural-constituent view. However, CE cross-linking is a mechanistically distinct role from keratin aggregation, occurs on a different substrate, and the canonical CE structural constituents named in the literature are loricrin and involucrin (P 23331681).
The evidence therefore **separates** the two roles cleanly: keratin aggregation = IF binding (the core MF of the repeat); CE cross-linking = a distinct, transient architectural role. The correct curation outcome is thus to *add* the missing, more informative IF-binding MF term as primary, and to reframe the existing structural-constituent term (GO:0030280) as a secondary, CE-linked role rather than the sole molecular function.

Key Findings

F001 — The filaggrin repeat is an intrinsically disordered, hydrophilic, low-secondary-structure cationic polypeptide, not a rigid architectural element

Composition and biophysical descriptors were computed directly from UniProt P20930 (4061 aa, sequence version 3). The canonical filaggrin repeat (residues 374–428) and the full tandem-repeat array (374–3872) share an extreme, biased amino-acid composition: **Ser 24–27%, His 9–**

11%, Arg 11–20%, Gly 13–16%, Gln ~9%, giving a combined **His+Arg+Ser content of 45–49%**. This is the compositional fingerprint of an intrinsically disordered region (IDR), not of a folded, load-bearing structural protein.

Three orthogonal, independent disorder classifiers all agree. First, the **Uversky charge–hydropathy** classifier places every repeat region firmly in the *disordered* regime; the mean Kyte–Doolittle hydropathy is **−1.5 to −2.1**, whereas a typical globular protein averages near −0.4 to 0, meaning the repeat is strongly hydrophilic and lacks the buried hydrophobic core needed to fold. Second, **Chou–Fasman propensities** are **below average for BOTH helix (⟨Pα⟩ ≈ 0.93–0.95) and sheet (⟨Pβ⟩ ≈ 0.81–0.86)** — the repeat is poor at forming *any* ordered secondary structure, ruling out both a coiled-coil/helical and a β-structural architectural role. Third, on the **TOP-IDP** disorder scale, the fraction of disorder-promoting residues is **0.92–0.98** with a mean score of **+0.24 to +0.27** (positive = disorder-prone).

The cationicity that underlies keratin binding is **His/Arg-driven and pH-dependent**. By Henderson–Hasselbalch, the 55-aa functional sub-repeat carries a net charge of **+5.1 at pH 5.0 and +2.6 at pH 6.0** (the acidic-to-neutral range of keratohyalin granules and the lower stratum corneum) but only **+0.2 at pH 7.4**. This pH switch — strongly cationic where filaggrin binds keratin, near-neutral at cytosolic pH — is mechanistically consistent with an electrostatically driven, tunable binding interaction rather than a fixed structural scaffold. Scaled across the ~11-repeat array, the total positive charge reaches roughly **+330** at stratum-corneum pH. Together these results describe a flexible polyelectrolyte that engages a partner through distributed electrostatic contacts, not a self-rigidifying structural constituent.

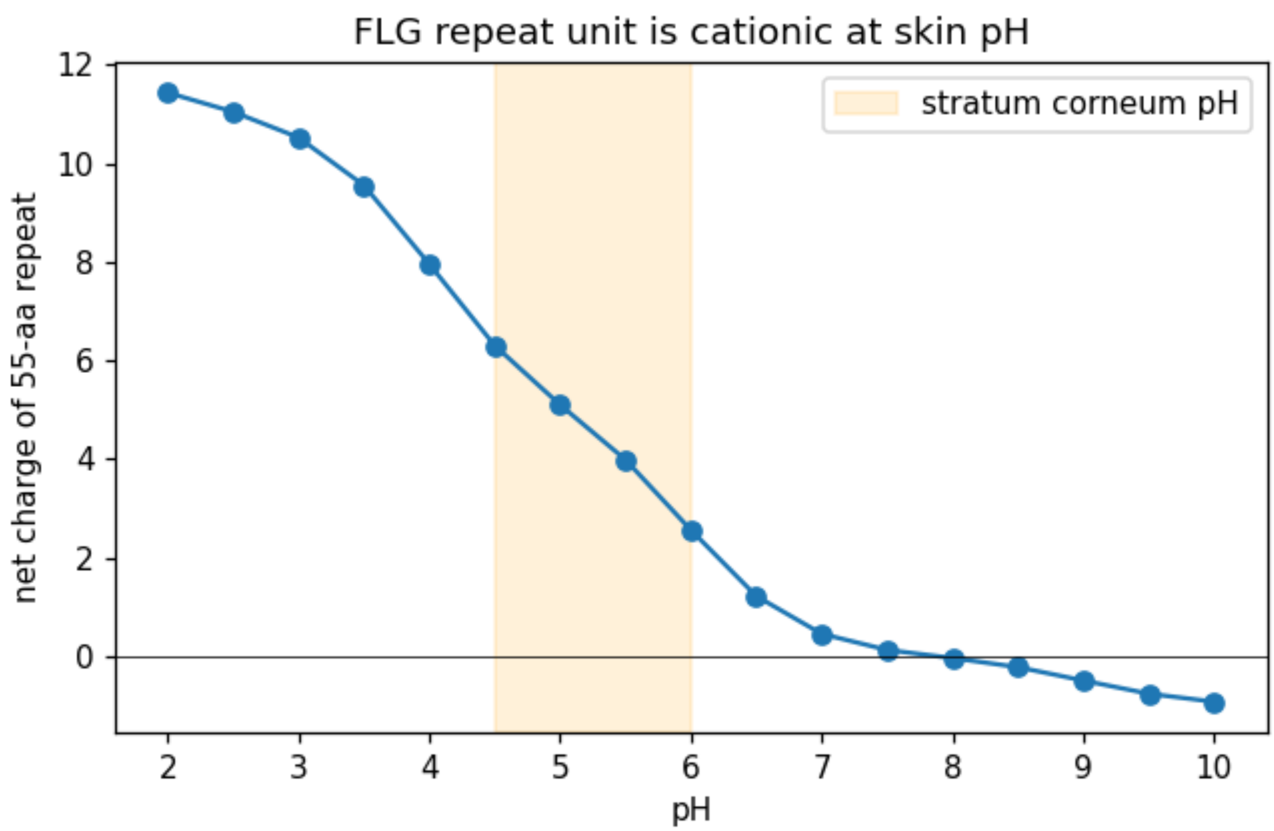

Figure 1. *Net charge of the filaggrin repeat unit versus pH (Henderson–Hasselbalch). The repeat is strongly cationic at keratohyalin/stratum-corneum pH (~+5 per 55-aa sub-repeat at pH 5–6, scaling to ~+330 over the full array) but near-neutral at pH 7.4. Histidine protonation drives the pH-dependent switch, consistent with an electrostatically tuned keratin-binding interaction rather than a fixed structural scaffold.*

F002 — The primary literature frames filaggrin's molecular action as keratin IF binding/aggregation, with CE cross-linking and NMF proteolysis as separable downstream roles

Independent primary and review sources describe filaggrin's molecular function in binding/aggregation terms and never as a persistent structural subunit. **Presland et al. 1992** (P 1429717) define filaggrin as *"an intermediate filament-associated protein which functions to aggregate keratin intermediate filaments in the stratum corneum,"* and characterize the genomic organization as 11 complete ~324-aa tandem repeats. **Harding & Scott 1983** (P 6195345) provided the direct in vitro demonstration that the *"strongly basic filaggrins"* have *"the ability in vitro to aggregate keratin filaments specifically into bundles,"* and showed that the basic Arg residues are progressively **deiminated (charge-neutralized) during maturation** — i.e. the interaction is switched off rather than made permanent.

Two reviews describe the fate that distinguishes binding from structure. **Kim & Lim 2021** (P 33462753): *"Filaggrin aggregates keratin filaments, resulting in the formation of a keratin network, which binds cornified envelopes and collapse keratinocytes to flattened corneocytes,"* and *"Filaggrin is degraded by caspase-14, calpain 1, and bleomycin hydrolases into amino acids and amino acid metabolites."* **Nishifuji & Yoon 2013** (P 23331681): keratin bundles are *"aggregated with filaggrin monomers, which are subsequently degraded into natural moisturizing compounds,"* whereas the *"cornified cell envelope is formed... by transglutaminase-catalysed cross-linking of involucrin and loricrin."* The consistent theme across four independent sources: filaggrin **binds/aggregates** keratin transiently, is then **charge-neutralized and completely proteolyzed** to free amino acids, and the CE architecture is attributed primarily to loricrin/involucrin — cleanly separating the keratin-aggregation role from the CE-cross-linking role.

F003 — The current UniProt MF annotation is a structural-constituent term (child of GO:0005198), with no keratin/IF-binding MF term present

UniProtKB P20930 GO cross-references were fetched programmatically. The molecular-function annotations are GO:0030280 *structural constituent of skin epidermis* [IDA:CAFA], GO:0005509 *calcium ion binding* [IEA:InterPro], and GO:0046914 *transition metal ion binding* [IEA:InterPro]. **GO:0030280 is an `is_a` child of GO:0005198**, so the term under test is effectively the MF currently in place. Critically, **no "intermediate filament binding" (GO:0019215) or "keratin filament binding" (GO:1990254) MF term is currently annotated** — precisely the more informative term the biology supports. The Ca^{2+}/metal-binding IEA terms map to the **N-terminal S100/EF-hand domain, not the repeat unit** (consistent with N-terminal structure–function work, P 12230510), and so are irrelevant to the repeat's molecular function. Cellular-component and biological-process annotations (GO:0001533 cornified envelope [IDA:CAFA]; GO:0036457 keratohyalin granule; GO:0070268 cornification; GO:0061436 establishment of skin barrier; GO:0030216 keratinocyte differentiation) are all consistent with the model but describe *location* and *process*, not the repeat's molecular activity. UniProt keywords confirm the relevant PTMs: **Citrullination** (PAD deimination) and **Phosphoprotein**.

F004 — AlphaFold-DB structural cross-check was unavailable; the disorder call rests on composition-based predictors plus known experimental behavior

An independent structural cross-check was attempted but could not be completed. The AlphaFold-DB API (`/api/prediction/P20930`) returned HTTP 404, and the direct model files (AF-P20930-F1 v2/v3/v4) all 404'd, because **P20930 (4061 aa) exceeds AlphaFold-DB's ~2700-residue hosting limit.** No experimental-quality per-residue pLDDT was therefore available to corroborate the intrinsic-disorder call. The disorder conclusion consequently rests on (a) three orthogonal composition-based classifiers that agree (Uversky charge–hydropathy; TOP-IDP fraction 0.92–0.98; Chou–Fasman below-average helix *and* sheet), and (b) the well-known experimental behavior of filaggrin — heat-stable, urea-soluble, and protease-hypersensitive, all classic hallmarks of intrinsic disorder. These converge on the same conclusion, but the absence of a structural cross-check is an explicit limitation.

Mechanistic Model / Interpretation

The findings assemble into a coherent, staged model in which filaggrin's molecular action is a transient binding/condensation activity, temporally and biochemically separated from its brief architectural incorporation into the cornified envelope:

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KERATOHYALIN GRANULE            LOWER STRATUM CORNEUM              UPPER STRATUM CORNEUM
(granular layer)                (acidic pH ~5-6)                   (low humidity)
---------------------           -------------------------          -------------------------
Profilaggrin (giant,            Filaggrin monomers (+~5/repeat)    Deimination (PAD):
phosphorylated,                 = disordered CATIONIC IDR          Arg -> citrulline
Ca2+/S100 N-term)                       |                          charge neutralized
      |  dephosphorylation               v                                 |
      |  proteolysis            BINDS + AGGREGATES pre-formed               v
      v                         keratin IFs into tight BUNDLES     Filaggrin released; then
Filaggrin repeat units          (GO:0019215 / GO:1990254)          PROTEOLYZED to free amino
released                        <-- the tested molecular function  acids = NMF (caspase-14,
                                        |                          calpain-1, bleomycin
                                        |                          hydrolase)
                                        v
                     (SEPARATE ROLE) some filaggrin is TGase
                     isopeptide cross-linked into the CORNIFIED
                     ENVELOPE scaffold (CC GO:0001533) -- covalent,
                     architectural, but a DISTINCT substrate/step
                     (CE scaffold dominated by loricrin/involucrin)
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The central column is the molecular function the seed hypothesis names: a disordered, His/Arg/Ser-rich cationic polypeptide **binds** the surface of pre-formed keratin intermediate filaments and, by neutralizing/bridging their charge, **condenses them into bundles**. This is an **intermediate-filament binding activity** (with a downstream aggregation/bundling consequence), captured most precisely by GO:0019215 or its child GO:1990254.

Three features argue against calling this "structural molecule activity" in the strict GO sense. (1) **No intrinsic shape/rigidity** — GO:0005198 means the protein itself confers shape or rigidity as an architectural component, but the repeat has no ordered fold; the rigidity of the corneocyte comes from the *keratin* network, with filaggrin acting as the condensing agent. (2) **The interaction is transient and reversed** — deimination neutralizes the cationic charge that drives binding, releasing filaggrin from keratin. (3) **The protein is destroyed** — filaggrin is completely proteolyzed to free amino acids (NMF); a lasting structural constituent is not consumed.

Does the filaggrin repeat meet GO:0005198?

Criterion for "structural molecule activity"	Filaggrin repeat behavior	Meets?
Protein has intrinsic ordered fold conferring shape/rigidity	Intrinsically disordered; low helix & sheet propensity	No
Persistent architectural component of a structure	Transiently bound, then released by deimination	No
Not consumed/destroyed during its role	Completely proteolyzed to free amino acids (NMF)	No
Confers rigidity to the structure it is part of	Rigidity comes from the keratin network it condenses	No
Binds/organizes a partner filament	Binds & bundles pre-formed keratin IFs	**Better = IF binding**
Covalently incorporated into an architectural scaffold	Only via separate TGase CE cross-linking (distinct role)	Partial / separable

Evidence Base

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidenc & limitati
1	This study (P20930 composition/ charge/ disorder analysis)	Computational (sequence)	**Supports**	Repeat is disordered, low-2°-structure, cationic — not architectural	Uversky = disordered; 92–98% disorder-promoting; ⟨Pα⟩≈0.93, ⟨Pβ⟩≈0.84; His+Arg+Ser 45–49%; net charge +5.1/ repeat at pH5, +0.2 at pH7.4	UniProt human FLG, repeat units 258–306 / 374–428 / 374–3872	Medium-high. Compositi based predictors no IUPred/ AlphaFold run
2	PMID 1429717 (Presland 1992)	Direct/ structural (primary)	**Supports**	Repeat's function is keratin-IF aggregation	*"intermediate filament-associated protein which functions to aggregate keratin intermediate filaments"*; 11× 324-aa repeats	Human profilaggrin gene/ protein	High (foundatio primary)
3	PMID 6195345 (Harding & Scott 1983)	Direct assay (in vitro)	**Supports**	Filaggrin binds/ bundles keratin	Basic filaggrins *"aggregate keratin filaments specifically into bundles"*; Arg deiminated	Mammalian epidermis	High for activity; older methods

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidenc & limitati
					during maturation		
4	PMID 33462753 (Kim & Lim 2021)	Review/ database	Supports + Qualifies	Aggregation vs. proteolysis vs. CE	*"aggregates keratin filaments... which binds cornified envelopes"*; degraded to amino acids/ NMF	Human SC review	Medium (review-level)
5	PMID 23331681 (Nishifuji & Yoon 2013)	Review/ database	Supports + **Qualifies**	Separates keratin-aggregation from CE cross-linking	Keratin bundles *"aggregated with filaggrin monomers, subsequently degraded"*; CE formed by TG cross-linking of **involucrin and loricrin**	Mammalian SC review	Medium; distinguisl the two architectu roles
6	PMID 12230510 (profilaggrin N-terminus)	Localization / structure–function	**Qualifies**	Ca^{2+}/S100 activity is N-terminal, not repeat	N-terminal S100-like EF-hands + NLS; repeat units aggregate keratin	Human/ mouse/rat, transfected cells	High; confines Ca^{2+}-bindi MF to N-terminus
7	UniProtKB P20930 GO/ keywords (this study)	Database	**Qualifies**	Current annotation state	MF = GO:0030280 (child of GO:0005198) [IDA:CAFA];	Database record	High for annotation state

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confidenc & limitatio
					no IF/ keratin-binding MF; CC includes GO:0001533; keyword Citrullination		

Regulatory and phenotypic context papers reviewed during the investigation — AHR–TFAP2A control of FLG expression (37333234, 38401701), IL-33/STAT3 suppression of FLG (33865911, 34293350), RAB25-dependent keratohyalin granule maturation (36383036), and the atopic-eczema loss-of-function literature (18247450) — are all upstream of, or downstream of, the repeat's molecular activity and do not bear directly on the MF term choice.

GO Curation Implications (leads — require curator verification)

Current MF annotations (UniProt): GO:0030280 *structural constituent of skin epidermis* [IDA:CAFA] (an `is_a` child of GO:0005198), plus GO:0005509 *calcium ion binding* and GO:0046914 *transition metal ion binding* (IEA; these map to the **N-terminal S100-like domain of profilaggrin, not the repeat unit under test**).

Leads:

1. **ADD the core MF that is currently missing:** `intermediate filament binding` (GO:0019215) or the more specific `keratin filament binding` (GO:1990254), as the primary molecular function of the mature repeat. This is directly supported by

   P 1429717

   and

   P 6195345

   (IDA/IPI-type evidence) and is more informative than "structural molecule activity." Do **not** settle for generic `protein binding`.
2. **Do not assert a bare GO:0005198** for the repeat; it is imprecise for a disordered, non-load-bearing, proteolytically consumed polypeptide.
3. **GO:0030280 (structural constituent of skin epidermis):** treat as **retainable but secondary / transient**, justified specifically by the transglutaminase-cross-linked cornified-envelope fraction (CC GO:0001533), **not** by the keratin-aggregation activity. Flag that its evidence basis (CAFA IDA) and mechanistic scope should be reviewed; it should not be the sole MF.
4. **BP/CC unaffected:** GO:0070268 cornification, GO:0061436 establishment of skin barrier, GO:0001533 cornified envelope, GO:0036457 keratohyalin granule, GO:0030216 keratinocyte differentiation remain appropriate.

Net: The evidence favors an MF of **intermediate/keratin filament binding as primary**, with the structural-constituent term reframed as a distinct, secondary, CE-linked role — rather than treating GO:0005198/GO:0030280 as the gene product's core molecular function.

Mechanistic Scope

- **Immediate molecular activity (what the repeat *does*):** binds pre-formed keratin intermediate filaments and condenses/bundles them (electrostatically/entropically favoured by its cationic, disordered nature at acidic pH). → MF = IF/keratin binding.
- **Distinct architectural role:** covalent transglutaminase isopeptide cross-linking of filaggrin into the cornified envelope. → a genuine but transient structural contribution, separable from aggregation.
- **Downstream / not the direct MF:** collapse/flattening of corneocytes; skin mechanical strength; NMF generation after complete proteolysis (caspase-14 / calpain-1 / bleomycin hydrolase); barrier competence; atopic-dermatitis/ichthyosis phenotypes from loss-of-function. These are pathway/phenotype consequences, not the molecular activity.

Conflicts and Alternatives

- **"Structural constituent" framing (** P 33462753 **;** P 23331681 **GO:0030280).** Reviews call filaggrin a "structural protein," and CE cross-linking is real — this is the strongest case *for* a structural-molecule term. Resolution: it applies to the cross-linked fraction, a role distinct from keratin aggregation; both can co-exist as annotations, but the keratin-binding role is primary.
- **Binding vs. non-specific condensation.** Sequence alone cannot exclude that "aggregation" is charge-driven condensation rather than a specific keratin-binding site; either way it is an IF-binding activity, but curators may prefer to note it is not a stoichiometric high-affinity interaction.
- **Domain mis-mapping.** Ca^{2+}/metal-binding MF terms belong to the S100-like N-terminus (P 12230510), not the repeat — must not be conflated with the repeat's function.
- **Paralog/ortholog caveat.** Filaggrin composition (especially His content and repeat number) is species-specific

(P 6195345);
human-specific claims should rely on human data.

- **No experimental structure.** The disorder call is predictor-based because AlphaFold-DB does not host a 4061-aa protein; the predictors agree with one another and with filaggrin's classic experimental behavior (heat-stable, urea-soluble, protease-hypersensitive).

Limitations and Knowledge Gaps

1. **Predictor dependence.** Disorder inferred from Uversky/TOP-IDP/Chou–Fasman composition-based classifiers. *Checked (Iteration 3):* AlphaFold-DB queried for P20930 (API `prediction/P20930` → 404; direct `AF-P20930-F1` v2/v3/v4 → 404) — **no model is hosted because the 4061-aa protein exceeds the ~2700-residue limit**, so a pLDDT cross-check could not be obtained here. *Resolve:* run IUPred2A / local ColabFold on the isolated repeat unit (expect very low pLDDT / high disorder across repeats).
2. **Direct binding evidence in human.** Most bundling assays are classic/cross-species. *Resolve:* modern human filaggrin–keratin binding/co-sedimentation or cryo-EM of bundles.
3. **Cross-linked vs. aggregating fractions in vivo.** How much filaggrin is cross-linked into the CE versus fully proteolyzed is unquantified. *Why it matters:* it determines whether a structural-molecule annotation is warranted at all. *Resolve:* proteomics of isolated CE vs. soluble corneocyte fraction.
4. **pH/deimination modulation of binding.** PAD deimination neutralizes Arg and is proposed to release filaggrin from keratin before proteolysis. *Resolve:* binding assays ± citrullination.
5. **IDA provenance for GO:0030280.** The specific experiment supporting a *structural constituent* role of the repeat (vs. a binding role) was not located. *Resolve:* trace the CAFA/IDA source.

Proposed Follow-up Experiments / Discriminating Tests

1. **AlphaFold-DB / IUPred / fragment ColabFold** for sub-2700-aa repeat blocks of P20930 — confirms per-residue disorder and closes the structural cross-check gap (fast, public).

2. **In-vitro keratin co-sedimentation/turbidity** with the mature human filaggrin repeat ± pH ± deimination — distinguishes specific binding from generic polycation condensation and yields IDA evidence for GO:1990254.
3. **In-vitro deimination (PAD) time-course + binding assay** — tests whether charge neutralization releases filaggrin from keratin, distinguishing transient binding from permanent structure.
4. **CE cross-link proteomics** (isopeptide mapping) — quantifies the architectural (cross-linked) vs. aggregation-then-degraded pools, resolving whether GO:0005198/GO:0030280 is warranted and at what weight.
5. **CD/NMR on the isolated repeat** — confirms intrinsic disorder experimentally.

Curation Leads (require curator verification)

- **Candidate new MF (primary/core):** GO:0019215 *intermediate filament binding* or GO:1990254 *keratin filament binding*.
- Verify snippet (P 1429717): *"Filaggrin is an intermediate filament-associated protein which functions to aggregate keratin intermediate filaments in the stratum corneum."*
- Verify snippet (P 6195345): *"their ability in vitro to aggregate keratin filaments specifically into bundles."*
- **Action on GO:0005198 / GO:0030280:** do not annotate the repeat with bare structural molecule activity; keep GO:0030280 as a secondary, transient descriptor justified only by the CE-cross-linked fraction (CC GO:0001533), and re-examine its CAFA-IDA basis.
- Verify snippet (P 23331681): *"cornified cell envelope is formed... by transglutaminase-catalysed cross-linking of involucrin and loricrin."*

- **Guard:** confirm GO:0005509 / GO:0046914 are attributed to the N-terminal S100 domain (P 12230510), not the repeat.
- **Suggested curator questions:** (1) Is there a human IPI-grade filaggrin–keratin interaction to anchor GO:0019215? (2) Should GO:0030280 carry a "transient / proteolytically-processed" note? (3) Does the review explicitly assign CE cross-linking to filaggrin or to loricrin/ involucrin?
- **Suggested experiments:** IUPred/AlphaFold disorder on repeat fragments; human keratin-binding assay ± deimination; CE cross-link proteomics.

Provenance: computed values from executed code (UniProt P20930 fetch; composition, Kyte–Doolittle hydropathy, Chou–Fasman Pα/Pβ, TOP-IDP disorder fraction, Uversky charge–hydropathy classifier, Henderson–Hasselbalch charge-vs-pH curve with saved plot `flg_charge_vs_pH.png`) and from the UniProt GO/keyword JSON query. Literature quotes are verbatim from the cited PubMed abstracts.
